A machine learning-based prediction model for adverse outcomes in women with preeclampsia

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Abstract

Background: **Preeclampsia is a major obstetrical disease that can increase maternal and fetal morbidity and mortality. Clinical guidelines have suggested several criteria for differentiation 'severe' from 'non-severe' preeclampsia to determine the need for indicated preterm delivery. However, these traditional criteria can be misleading because 'non-severe' preeclampsia based on these criteria may rapidly progress to 'severe' disease in clinical circumstances. In the current study, we developed machine learning based prediction model to accurately predict the need for early preterm birth (<34 weeks).**

Methods: **This prospective cohort study included singleton pregnant women who received antenatal care at Seoul National University Hospital. The initial dataset consists of 315 patients with 47 clinical variables at the time of admission. Primary outcome was delivery before 34 weeks of gestation. The prediction model was constructed using all applicable predictors and an ensemble approach, which aggregates eight machine learning (ML) models. The ensemble prediction model was evaluated using both augmented external validation and independent external validation.**

Results: **Using predictive Ensemble ML modelling, we developed prediction model for predicting delivery before 34 weeks of gestation with the use of clinical variables at admission. The prediction model showed good discrimination (area under the ROC curve (AUROC) = 0.86 (95% CI 0.75-0.98)) with 81% of sensitivity and 83% of specificity, whereas traditional criteria (severe vs non-severe PE) showed 84% of sensitivity and 48% of specificity (p = 0.72 against sensitivity and <0.0001 against specificity). Among the clinical variables included in the prediction model, alkaline phosphatase, systolic blood pressure and fibrinogen were the top three influential variables. External validation using Boramae Medical Center also showed good discrimination (AUROC 0.78, 95% CI 0.71-0.84).**

Conclusions: **The combined use of maternal factors and common antenatal laboratory data could effectively predict adverse pregnancy outcomes using machine learning algorithms**

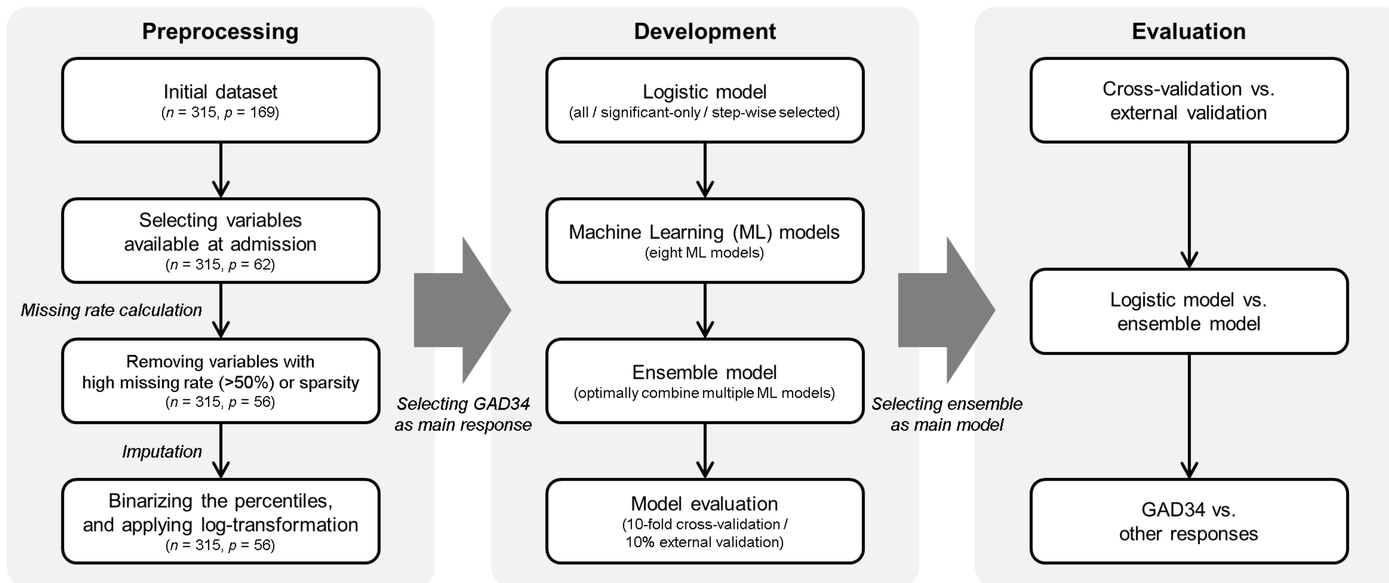
Materials & Methods

Study design: Prospective cohort study

Study population: Singleton pregnant women with preeclampsia (SNUH)

Variable : 47 clinical variable at the time of admission

Primary outcome : delivery before 34 weeks of gestation



Results

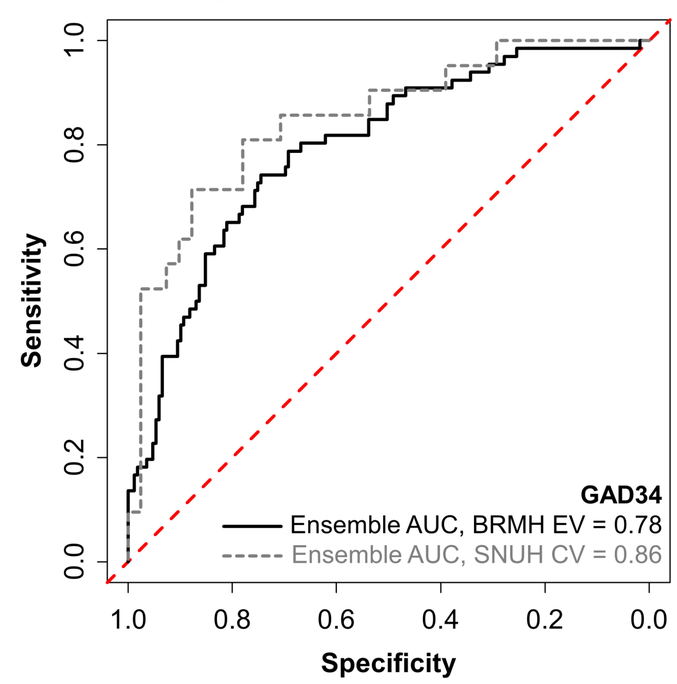


Figure 1-. The performance of prediction model in internal test and external validation set

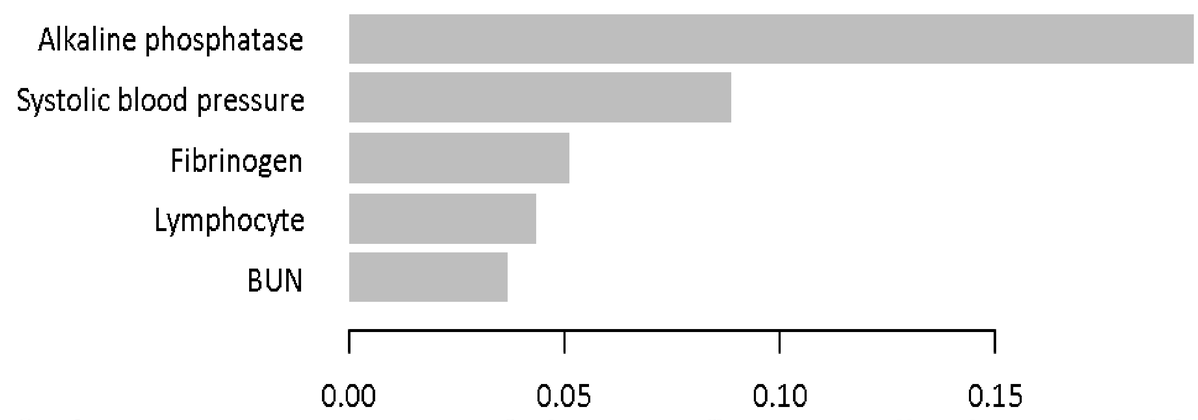


Figure 2. Importance scores of the top five contributing variables to the prediction model.

Conclusion

The combined use of maternal factors and common antenatal laboratory data could effectively predict adverse pregnancy outcomes using machine learning algorithms